

Chapter 18 - Nixtamalized Maize Flour By-product as a Source of Health-Promoting Ferulated Arabinoxylans (AX)

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Abstract

Maize is a major source of food in Mexico, especially for low-income populations. To improve its [nutritional value](#), maize kernel is exposed to an alkali-treatment called [nixtamalization](#). However, this process generates large volumes of alkaline wastewater (*nejayote*) with a high concentration of organic matter, especially nonstarch [polysaccharides](#) like [arabinoxylans](#) (AX), which are part of the dietary fiber (DF) intake of humans. AX selectively enhances beneficial [colonic microbiota](#) growth, favoring good health in the host. This polysaccharide promotes [prebiotic](#), antioxidant, antitumoral, and immunomodulatory activities. Furthermore, AX can form covalent gels with potential applications as encapsulation agents in the food, pharmaceutical, and biomedical industries. This chapter discusses nejayote as a source of AX that presents good health promotional characteristics.